Decision Tree Classification: Effect of Tree Depth on Model Performance

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1. Introduction

Decision Tree is a supervised machine learning algorithm used for classification and prediction tasks. It works by splitting data into smaller groups based on feature values. Each split represents a decision, and by following these decisions step by step, the model reaches a final prediction.

Decision trees are widely used in real-world applications such as medical diagnosis, fraud detection, and customer segmentation. Their ability to provide clear decision rules makes them useful in situations where interpretability is important. In machine learning, understanding how model complexity affects performance is essential, and decision trees provide a simple yet powerful way to study this concept.

One of the main advantages of decision trees is that they are easy to understand and visualize. However, they also have a common problem called **overfitting**, where the model becomes too complex and starts memorizing the training data instead of learning general patterns.

In this tutorial, the focus is on understanding how the depth of a decision tree affects its performance and how it can lead to overfitting.

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2. Objective

The objectives of this tutorial are:

- To understand how a decision tree works
- To compare a small tree and a large tree
- To analyse the effect of tree depth on performance
- To demonstrate overfitting using both visual and numerical results

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3. Working of Decision Tree

3.1 How Decision Trees Work

A decision tree works by recursively splitting the dataset into smaller subsets based on feature values. At each step, the algorithm selects the best feature using criteria such as Gini impurity or entropy. The goal is to create splits that result in the most homogeneous groups.

The process continues until a stopping condition is met, such as reaching maximum depth or having too few samples in a node. The final nodes are called leaf nodes, where predictions are made.

This hierarchical structure makes decision trees easy to interpret and visualize.

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4. Dataset

A synthetic dataset was created using the make_classification function from the Scikit-learn library.

- Number of samples: 600
- Number of features: 2
- Number of classes: 2

This dataset was chosen because it allows easy visualization of how the model splits the data. A small amount of noise was also added to make the data more realistic instead of perfectly separable.

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5. Methodology

The use of a synthetic dataset allows better control over the distribution and noise levels. This helps in clearly observing how the model behaves under different complexities. The training and testing split ensures that the model is evaluated on unseen data, which is essential for measuring generalization performance.

5.1 Data Generation

The dataset was generated with two features so that it can be plotted and understood visually. This makes it easier to see how the decision tree is working.

5.2 Train-Test Split

The dataset was divided into:

- 70% training data
- 30% testing data

The training data is used to train the model, and the testing data is used to evaluate how well the model performs on unseen data.

The implementation was done using Python with the Scikit-learn library for building and evaluating the decision tree models.

5.3 Model Training

Two decision tree models were trained:

- **Small Tree** with limited depth (max_depth = 3)
- **Big Tree** with no depth limit

The decision tree uses the **Gini index** to choose the best split at each step.

5.4 Evaluation Methods

The models were evaluated using:

- Accuracy score
- Confusion matrix
- Visual plots such as decision boundaries and graphs

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6. Results and Analysis

The results clearly indicate that increasing the depth of the decision tree increases its ability to fit the training data. However, this also leads to capturing noise and irrelevant patterns. As a result, the model performs worse on test data, demonstrating overfitting.

On the other hand, a shallow tree may not capture all patterns, leading to underfitting. Therefore, selecting an optimal depth is critical for achieving a balance between bias and variance.

6.1 Dataset Visualization

The dataset plot shows two classes with some overlap. This makes the classification problem more realistic and slightly challenging.

All plots use clear labels and colours to make them easy to understand.

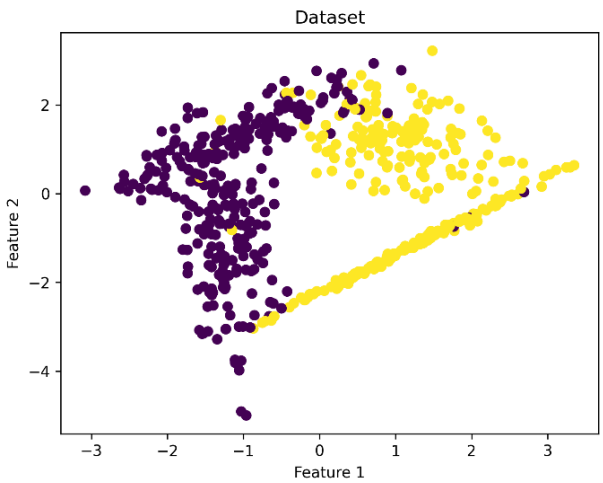

Figure 1: Distribution of the synthetic dataset showing two classes with slight overlap.

6.2 Decision Boundary

The decision boundary plots show how each model separates the data:

- The **small tree** creates simple and smooth decision regions
- The **big tree** creates more complex and irregular regions

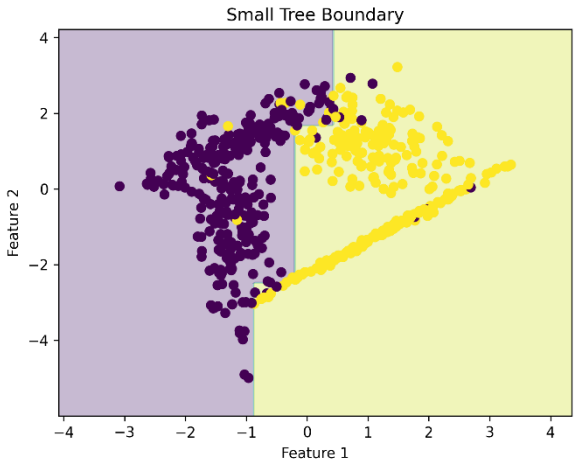

Figure 2: Decision boundary created by the small decision tree (max depth = 3), showing a simple split.

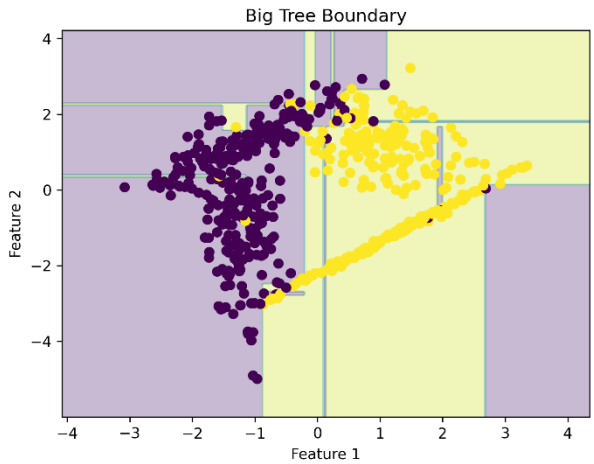

Figure 3: Decision boundary created by the large decision tree, showing more complex and irregular regions.

This shows that increasing tree depth increases the complexity of the model.

6.3 Accuracy Comparison

- Small Tree Accuracy: approximately 0.93
- Big Tree Accuracy: approximately 0.92

Even though the big tree is more complex, it does not perform better on test data.

6.4 Confusion Matrix

The confusion matrix shows the number of correct and incorrect predictions:

- The small tree performs slightly better
- The big tree makes more mistakes on unseen data

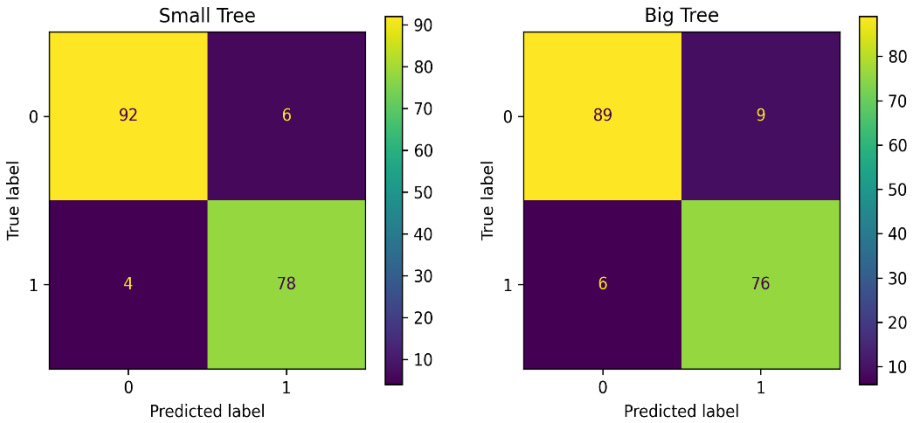

Figure 4: Confusion matrices for small and large trees showing prediction performance on test data.

6.5 Feature Importance

The feature importance results show that:

- Feature 1 has a higher importance
- Feature 2 has lower importance

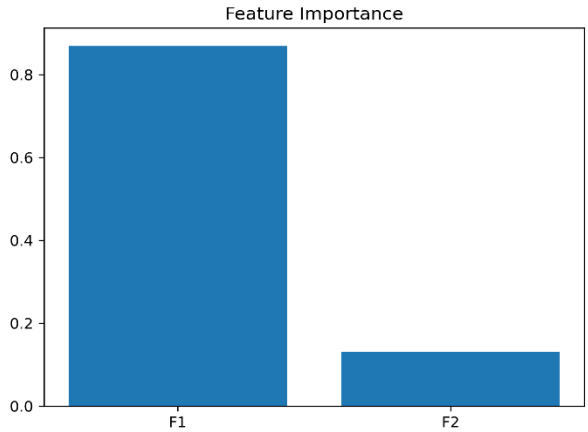

Figure 5: Feature importance values indicating the contribution of each feature to the model.

This means the model depends more on one feature when making decisions.

6.6 Decision Tree Structure

The tree diagram shows how decisions are made step by step:

- The top nodes split the data into large groups
- The lower nodes make more detailed decisions

  As the dept increases, the tree becomes more detailed and complex

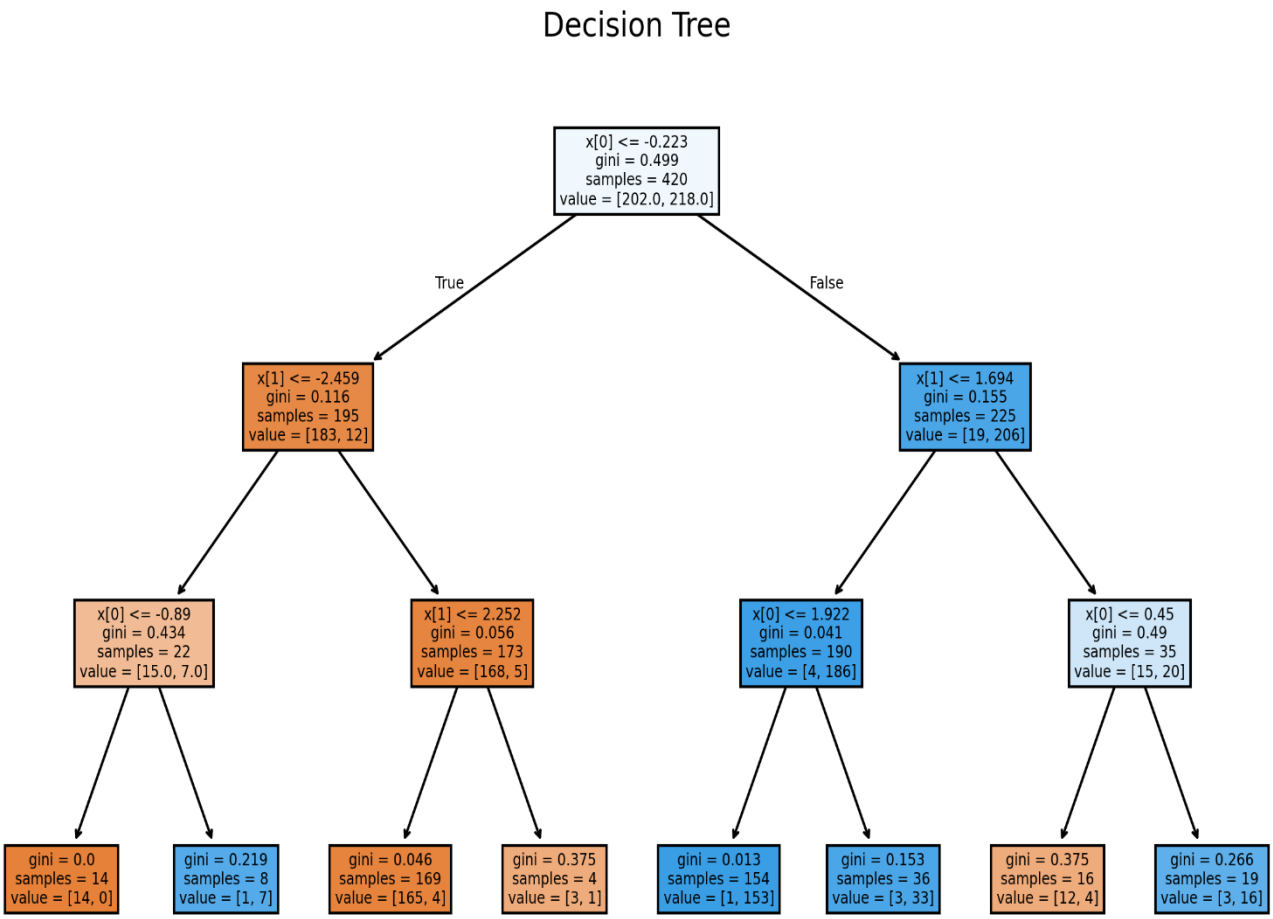

Figure 6:Structure of the decision tree illustrating how the data is split at each step

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6.7 Overfitting Analysis

The overfitting graph shows how accuracy changes with tree depth:

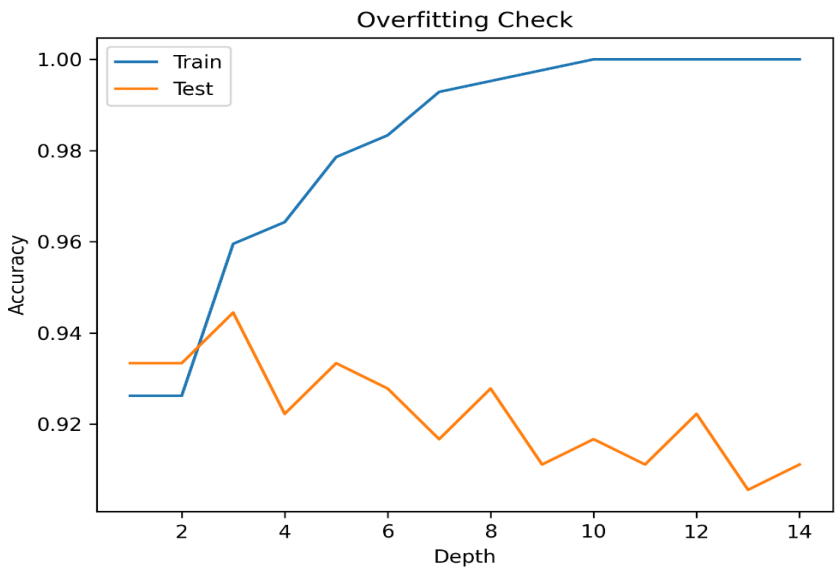

Figure 7: Training and testing accuracy across different tree depths showing overfitting behaviour.

- Training accuracy keeps increasing and reaches 100%
- Testing accuracy increases at first but then decreases

The deep tree creates many splits that perfectly fit training data but fail to generalize, which is a sign of high variance.

This clearly shows that a very deep tree leads to overfitting.

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7. Discussion

From the results, it is clear that:

- A shallow tree may be too simple and miss patterns (underfitting)
- A deep tree becomes too complex and memorizes data (overfitting)
- The best model is somewhere in between

This shows the importance of controlling tree depth to get better performance.

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8. Advantages and Limitations

Decision trees have several advantages. They are easy to understand, require little data preprocessing, and can handle both numerical and categorical data. They also provide clear decision rules that are easy to interpret.

However, they also have limitations. Decision trees are prone to overfitting, especially when they become too deep. They are also sensitive to small changes in data, which can result in different tree structures. This makes them less stable compared to other models.

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9. Conclusion

In this tutorial, we studied decision trees and how their depth affects performance.

Key points:

- Decision trees are simple and easy to understand
- Increasing depth increases model complexity
- Overfitting happens when the tree becomes too deep
- Choosing the right depth is important for good results

Overall, this study highlights the importance of balancing model complexity to achieve better accuracy.

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10. References

- Scikit-learn Documentation: https://scikit-learn.org/stable/modules/tree.html
- Géron, A. (2019). *Hands-On Machine Learning with Scikit-Learn*
- Towards Data Science – Decision Tree Explained

GitHub Repository

LINK: https://github.com/tamilvendan066-lab/decision-tree-depth-analysis.git